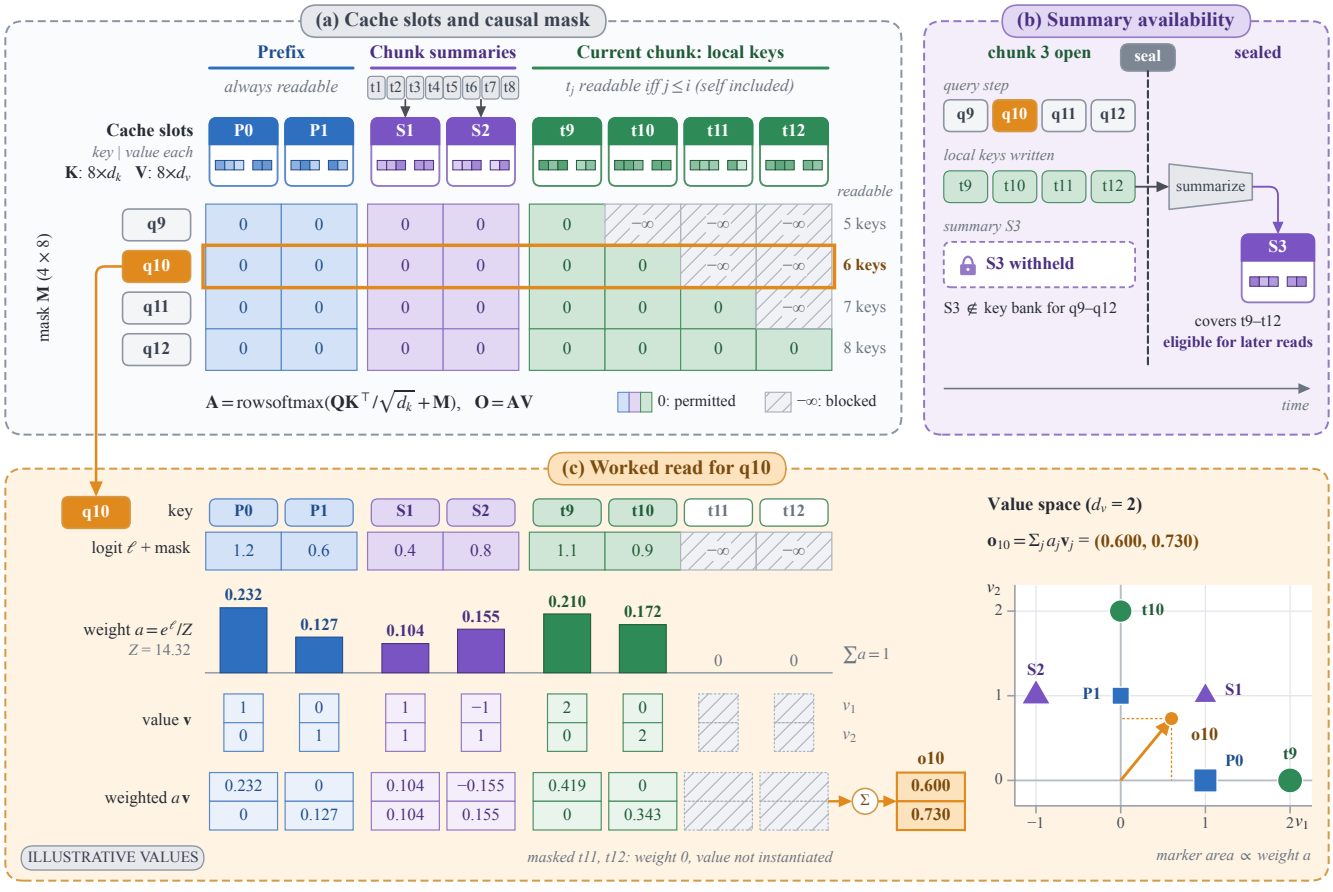

(a) Cache slots and causal mask
Prefix
always readable
Chunk summaries
t1 t2 t3 t4 t5 t6 t7 t8
Current chunk: local keys
t_j readable iff $j \le i$ (self included)
Cache slots
key | value each
K: 8×d_k V: 8×d_v
P0 P1 S1 S2 t9 t10 t11 t12
readable
mask M (4 × 8)
q9 0 0 0 0 0 −∞ −∞ −∞ 5 keys
q10 0 0 0 0 0 0 −∞ −∞ 6 keys
q11 0 0 0 0 0 0 0 −∞ 7 keys
q12 0 0 0 0 0 0 0 0 8 keys
$\mathbf{A} = \mathrm{rowsoftmax}(\mathbf{QK}^\top/\sqrt{d_k} + \mathbf{M}), \quad \mathbf{O} = \mathbf{AV}$
0: permitted
−∞: blocked
(b) Summary availability
chunk 3 open
seal
sealed
query step
q9 q10 q11 q12
local keys written
t9 t10 t11 t12
summarize
summary S3
S3 withheld
S3 ∉ key bank for q9–q12
S3
covers t9–t12
eligible for later reads
time
(c) Worked read for q10
q10
key
P0 P1 S1 S2 t9 t10 t11 t12
logit ℓ + mask
1.2 0.6 0.4 0.8 1.1 0.9 −∞ −∞
weight $a = e^{\ell}/Z$
$Z = 14.32$
0.232 0.127 0.104 0.155 0.210 0.172 0 0
$\sum a = 1$
value v
1 0 / 0 1
0 1 / 1 1
−1 1 / 2 0
0 2
v_1
v_2
weighted $a\mathbf{v}$
0.232 0 / 0 0.127
0.104 −0.155 / 0.104 0.155
0.419 0 / 0 0.343
Σ
o10
0.600
0.730
Value space (d_v = 2)
$\mathbf{o}_{10} = \Sigma_j a_j \mathbf{v}_j$ = (0.600, 0.730)
v_2
t10
S2
P1
S1
o10
P0
t9
v_1
ILLUSTRATIVE VALUES
masked t11, t12: weight 0, value not instantiated
marker area ∝ weight a